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Rocks and racism? How geologists created and perpetuated a narrative of prejudice

Kathryn Yusoff sparked a culture war with her latest book, suggesting slavery and white supremacy informed the work of geology's founding fathers. Here, she and other experts suggest that attitudes have changed little since

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Sun 16 Mar 2025 09.00 CET

It was at the London Library at St James's Square, surrounded by the shiny offices of geological extraction companies including BP and Rio Tinto, that Kathryn Yusoff discovered a deep link between geology and racism. It was 2017, and the professor of inhuman geography from Queen Mary University of London was doing research for a book about the history of geology. Little did she know that her niche, archival discoveries, which led to a 600-page tome on race and geoscience seven years later, would put her on the very faultline of a culture war.

The academic book, called *Geologic Life*, was published in 2024. It soon attracted the attention of the conservative higher education publication the College Fix, which said that Yusoff's writing "suggests even rocks have been corrupted in white supremacists' schemes". A minor eruption followed, with articles in many rightwing newspapers including the *Daily Mail* and *Telegraph*, featuring quotes such as "geology is no more racist than fish'n'chips". "The articles were grossly inaccurate and deliberately mischaracterised the book in inflammatory and abusive language," says Yusoff.

We tend to think of biology, and the theory of evolution in particular, as the scientific inspiration for debunked racist ideas, such as eugenics. But geology is older than biology. "Charles Darwin actually studied geology," explains Yusoff. "Geology was the discipline in which race was historically made, as both a set of ideas and a material transformation of the Earth that was racialised."

Geology was intimately linked to colonialism, developing rapidly from the 17th to the 19th centuries. Yusoff says its rise was propelled by an earthquake in 1755 that practically destroyed the city of Lisbon, a centre of the slave trade. The event created a "geotrauma" for Europeans, she says. "The shock of the Lisbon earthquake started to generate enlightened thought," she argues. "Science replaced the idea of God as the explanation for what was going on when the Earth shook."

Geology provided a way to systematically classify and map the Earth and its history. It became *the* science of the New World, according to Yusoff, with geological surveys ultimately helping settlers find the best land. Geological knowledge helped build plantations and mines, extracting valuable resources while creating a geotrauma for black and Indigenous populations. Their knowledge of the Earth was essentially erased and excluded.

This is only part of the story, however. Yusoff's book argues that the scientific ideas themselves at this time were also racist and sought to legitimise white supremacy – long before such ideas emerged from biology.

Early geology mainly focused on determining local formations of strata (layers of rock) for commercial mining. But between the 1760s and 1820s, it morphed into a set of theories about the Earth and the study of life across time, incorporating palaeontology. "At this time, geology was about looking at people's bodies and trying to put them in time," says Yusoff. There were few trained geologists in colonial posts, explains Pratik Chakrabarti, a historian of science at the University of Houston. "A lot of them also did anthropology – and they used similar metaphors, such as strata, to talk about both," he says. Fossils were therefore often compared with living populations.

> *There was a narrative that Indigenous people would go extinct and that black people needed to be guided by the hand of whiteness*
> Kathryn Yusoff

This created a vertical view of time, with older fossils found deeper underground. The discovery that vertebrate life had become increasingly more diverse and complex over time also gave the illusion of progress though the strata. Yusoff says Indigenous and black people were seen as a link between extinct species and white "modern" people. They were part of the past, in the strata, while white Europeans imagined themselves as representing the present and the future - standing firmly on the surface. "There was a constant narrative that Indigenous people would go extinct and that black people were children who needed to be guided by the hand of whiteness," says Yusoff.

Chakrabarti has come to a similar conclusion in his research - showing it was a global problem, including in 19th-century India. "This attempt to compare Indigenous populations to prehistoric humans was a racist agenda, assuming these people hadn't changed in thousands of years," he says. "The term that was used was 'living fossils'."

Saartjie Baartman, who was exhibited in early 19th-century London, drew the attention of pioneering naturalists Geoffroy Saint-Hilaire and the Cuvier brothers. Photograph: Eraza Collection/Alamy

In the 1820s, the prominent palaeontologists Frédéric Cuvier and Geoffroy Saint-Hilaire published their studies of the natural history of mammals. Tellingly, the only human featured was a black woman called Saartjie Baartman, displayed between a deer and a langur monkey. Cuvier came across Baartman at a ball, where she was exhibited as a curiosity, and requested a private viewing. Cuvier's palaeontologist brother, Georges, even tried to bribe her to show him her genitals for "scientific purposes". Yusoff argues this was because he wanted to prove his theory that more "primitive" mammals had larger sexual organs and higher sex drive. Baartman apparently refused, but after her death, George Cuvier claimed her remains for science from the police, dissected her and published detailed diagrams of her labia and other body parts anyway, which were later reproduced by geologists including Darwin.

There's no shortage of examples of racist views among geologists in *Geologic Life*. Charles Lyell, a founding father of geology, had close ties to the enslaver James Hamilton Couper, and praised him for his skills. "The slaves identify themselves with their master, and their sense of their own importance rises with his success in life," he said in 1846. Louis Agassiz, another pioneering geologist, wrote: "The Indian woman… rarely has the feminine delicacy of higher womanhood."

These ideas persisted even after the emancipation of enslaved people. In the American south and South Africa, many black people who got arrested for minor offences were forced to work in mines - quite literally placed underground, below the white supervisors on the surface.

Mike Benton, a palaeontologist at the University of Bristol, says there is no doubt geology benefited from colonialism and that many geologists at the time were racist. "It's absolutely appropriate and important that these things are said," he argues. He also agrees that "the p word" - progress - could have exacerbated racist thinking.

But Benton believes it's important to keep in mind that prejudice ultimately comes out of interpreting the science rather than the science itself. He says it's "a bigger leap" to suggest the sequencing of rocks in time directly influenced sociological or political ideas. Geologists at the time ultimately had a poor understanding of concepts such as time and evolution, he argues, and they differed widely in their opinions. Some, including Lyell, viewed time as cyclical rather than linear, according to Benton, meaning he may not have subscribed to views of progress in the Earth's geological record.

Graham Shields, a geologist from University College London, is also sceptical about the link between strata and racism. "Without the theory of evolution, I can't see how they would have thought of strata in an evolutionary sense," he says.

Munira Raji disagrees. Originally from Nigeria, she is now a geologist at the University of Plymouth, and she has researched the links between her subject and colonial history. "I've seen colonial reports, and the way they describe my people, compared with some other local populations, is definitely as part of the strata - with some groups seen as smarter than others," she says. "And that's how Nigerians still see major ethnic groups today."

acist legacies of geology persist today, argues Yusoff. Rocks are still a source of power. In 2020, Rio Tinto destroyed a 46,000-year-old First Nations heritage site of

R rock shelters in Australia to expand an iron ore mine. The parents of some of the most powerful people on Earth, including tech entrepreneurs Elon Musk and Peter Thiel, have been linked to mines in sourthern Africa.

A protester outside Rio Tinto's offices in Perth, Australia in June 2020 following the destruction of ancient Indigenous rock shelters by the mining company. Photograph: Richard Wainwright/AP

What's more, although scientific racism has been debunked, it still lives on in other spheres of society. "Jair Bolsonaro [former president of Brazil] was always using the term 'stone age people' to refer to Indigenous people while trying to take their forests for mining," says Yusoff.

How we ended up in the Anthropocene, an era marked by human activity, also has an impact on how we deal with the fallout, including climate change, says Yusoff. "What do environmental reparations look like to allow livable futures for people who have been impacted through geo-traumatic pasts?" she asks. "Impacts from colonial mining are still there, with marginalised communities living with toxic residues and poor water quality today."

Raji says the legacy of colonialism also continues to influence modern geology practices. "Many geologists still do 'parachute science', where they go to Africa, take some samples and fly back. They then publish their findings without acknowledging local geological knowledge or experts." Another problem, she says, is that Indigenous knowledge gets written out of history books. For example, historical archives state that colonial geoscientists "discovered" coal in Nigeria. But local populations were already using it.

Benton says the UK may need to share or hand over collections of rocks and fossils to countries they were taken from. “It would be perfectly reasonable that a museum in another country, one in Africa for example, might like to have some collections,” he says.

Shields agrees that geologists should be aware of the discipline’s colonial past but also says he’s not sure how relevant all of this is today. “For me, geology is just about increasing knowledge. When I started working in China, some western scientists would consider the local, Chinese scientists as mere tour guides, which was a patronising attitude. But at the same time, you want to collaborate with the best people. As China has moved forward technologically, they produce so much great science that we struggle to keep up.”

Since the discovery that Europeans have significant amounts of Neanderthal DNA, evidence has emerged of how advanced they really were… Photograph: Action Press/Rex Features

Many people view decolonisation as "anti-scientific", says Yusoff. Chakrabarti argues such views, sometimes held by scientists themselves, are a defence mechanism, and "they better grow out of it". As he mentions in his forthcoming book *Science As a White Epistemology*, 19th-century scientists similarly dismissed concerns by animal rights activists as anti-scientific. "Now it is an integral part of science to do ethical checks," says Chakrabarti. "Ethics can make science richer, so why not tackle the question of race?"

Decolonisation "does sometimes make science better", argues Yusoff. Benton highlights the case of Piltdown Man, often dubbed the "first Englishman". In 1912, amateur archaeologist Charles Dawson said he had unearthed remains of the skull of an unknown human species in the UK, which he claimed was a missing link between early apes and humans. It was immediately compared to Indigenous people, with geologist Arthur Smith Woodward suggesting the brain size "equals that of some of the lowest skulls of the existing Australians". Despite criticism from scientists in other countries, it took until 1953 for the specimen to be uncovered as a fraud - possibly partly because of national pride and racist legacy.

Perhaps race was also a factor in the recent revival of Neanderthals. Neanderthal remains, first discovered in 1856, were initially compared to Aboriginal Australians. This species was also long considered inferior to modern humans. But as Angela Saini points out in her book *Superior: The Return of Race Science*, this changed in 2014 when DNA sequencing technology revealed that Europeans have a significant proportion of Neanderthal DNA, and much more so than black Africans, for example. Since then, researchers have started uncovering that the Neanderthals were actually rather advanced - capable of art and tradition.

Either way, Yusoff is hoping her book can help shift the debate about racism. "*Geologic Life* argues that a shift is necessary in the discourse around race from ideology and identity politics into discussion about material relations between people and environments," she says. "The extraction of geological materials ultimately sustains forms of power."

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